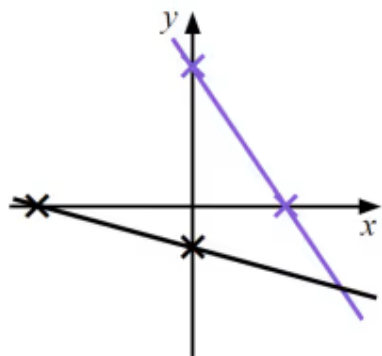




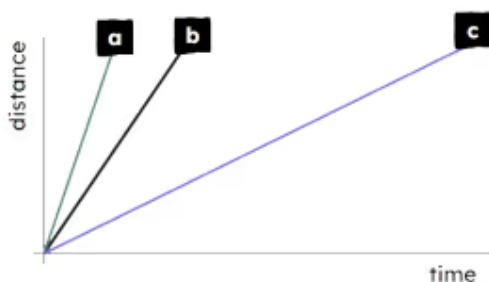
Problem solving with graphical representations

- 1 The y- intercept is the point where a line or curve meets the y-axis. Fill in the blank
- 2 Four pupils are solving a pair of simultaneous equations. The diagram shows the graphs of the equations they are solving. Which pupils are definitely incorrect? Tick **3** correct answers



- ☒ Alex says $x = 2, y = 3$
- ☒ Izzy says $x = -2, y = -3$
- ☐ Jun says $x = 2, y = -3$
- ☒ Laura says $x = -2, y = 3$

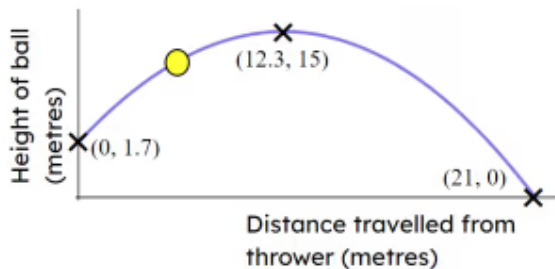
- 3 The graph models the time taken for three people to run 100 metres. Which line represents the fastest runner? Tick **1** correct answer



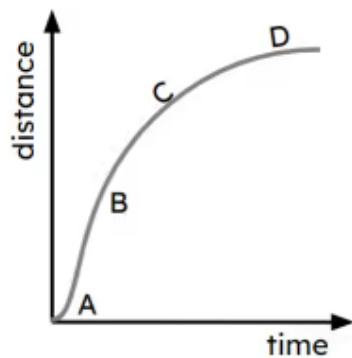
- ☒ Line a
- ☐ Line b
- ☐ Line c



- 4 The parabola is used to model the throw of a tennis ball. The ball reaches a height of 10 metres 2 times. Fill in the blank



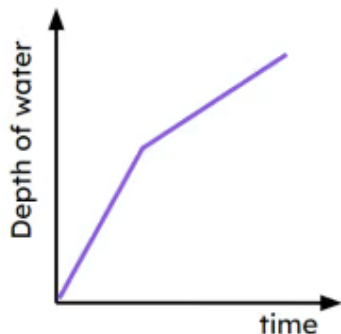
- 5 The graphs models Laura running a 100 metre race. The graph covers a distance of 110 metres. Match each section of the graph to the correct statement. Write the correct letter in each box



a	A
b	B
c	C
d	D

c	Laura is slowing down
b	Laura is running fast
d	Laura has nearly stopped
a	Laura is accelerating

- 6 A container is filled with water at a constant rate. The graph shows the depth of water in a container as it is filled. Which of these could be a cross section of the container?
Tick **1** correct answer


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